Swaroop Kumar Palai

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RESEARCH INTERESTS

Quantum Optics, Nano & Microfabrication, Metrology, 2D materials, Optoelectronics, Optical characterisation, Photonic and Optical engineering, Electrical instrumentation.

EXPERIENCES

Post-Doctoral Projects

1. Quantum-emitters-embedded dielectric light-emitting metasurfaces (Q-META)
CEMES & LPCNO, Toulouse, France *2025 – 2026*

- Controlling and enhancing emission properties of quantum emitters (NV centers in nanodiamonds) using 2D optical dielectric metasurfaces hosting Bound In the Continuum States.
- Deterministic positioning of nanodiamonds in the near field of dielectric arrays using directed assembly via AFM nanoxerography.
- Optical characterization at the single photon scale including time-resolved confocal microscopy and photon correlation.

2. Plasmonic Optical Memristive Memory of an Addressable pRocessing Device (POMMARD)
CEMES & LPCNO, Toulouse, France *2026 – Present*

- Designing nanofabrication workflows and developing deterministic sample preparation protocols for directed assembly of NV centers.
- Optimizing AFM nanoxerography surface charge patterning and positioning yields for single nanodiamonds near plasmonic Double Hexagon (DH) cavities.
- Establishing hybrid integration strategies of local quantum photon sources with on-chip plasmonic memristive architectures for interconnect-free optical logic units.

Doctoral Projects

1. Fabrication of 2D semiconducting heterostructures and improving their optical properties using AFM ironing.
CNRS-LNCMI, Toulouse *2021 – 2024*

- Developed a novel fabrication technique for TMDC heterostructures and used AFM to improve surface and optical characteristics of the sample.
- Built a setup for micro-photoluminescence (PL), absorption, non-linear spectroscopy, two-photon spectroscopy measurements at cryogenic temperatures (4K).

2. Construction and optimising flake probe for magneto-optical measurements.
CNRS-LNCMI, Toulouse *2021 – 2024*

- Building the flake probe fiber optical setup to measure at 70T.
- Coupling of fiber and optimsing the setup to minimise the losses.
- Calibrating the RTDs to give us temperature reading at 2K.
- Micro soldering of components like pickup coil and RTDs.

3. Project- Modulating atomic reconstruction on MoS_2/WS_2 using contact mode AFM
Nenovision s.r.o, Brno, Czech Republic *2024*

- Optimising the contact mode AFM measurements.
- Correlation spectroscopy utilising SEM and AFM using Litescope 2.5 in a cleanroom environment.

4. Studying the charge transfer in WSe_2-BAPI hybrid heterostructure
Politecnico di Milano, Milan, Italy *2023*

- Building the setup for polarised pump-probe measurements.
- Conducted pump-probe measurements on encapsulated WSe_2 monolayer to understand Dexter coupling.
- Conducted transient absorption spectroscopy on WSe_2-BAPI heterostructure to illustrate the charge transfer.

5. Building fibre coupling and transfer setup
Politechnica Wroclawska, Wroclaw, Poland *2023*

- Exfoliation of perovskite crystals.
- Optimising the setup to transfer the monolayer flakes to mono-mode fibres.

6. Fabricating TMDC Heterostructures and optimising transfer technique
ICMM - CSIC, Madrid, Spain *2022*

- Exfoliation of TMDC crystals and fabrication of heterostructures using nail polish.
- Conducted AFM ironing as a post-processing to improve interface quality.
- Conducted comprehensive statistical analysis and explored spectral parameters of TMDC heterostructures through AFM ironing.

Pre-Doctoral Projects

1. Flexible Quantum Light Emitting Devices from Van der Waal layered materials
IIT Madras, Chennai, India *2020 – 2021*

- Designing and optimisation of CVD furnaces and managing technical tenders.
- Characterizing flakes using OM, Raman and AFM.

2. Nanofabrication of flexible nanogaps
Seoul National University, Seoul, South Korea *2019*

- Specialised in nanofabrication with expertise in optimisation, wafer patterning, UV-lithography, (dry/wet) Etching, Lift-off, Atomic Layer Deposition (ALD) and nanofabrication.
- Worked in clean room environments fabricating the nanoantennas and topology-changing metasurfaces.

3. Studying the effect of pH on the dynamics of PAA using FRAP
NIT Rourkela, Rourkela, India *2019*

- Employed Fluorescence Recovery technique to understand the dynamics of the PAA.

EDUCATION

Doctoral Researcher in Material Science *Toulouse, FRANCE*
CNRS- LNCMI, Paul Sabatier University Toulouse III Sept. 2021 – Oct. 2024

Integrated Bachelors-Masters in Physics and Astronomy *Rourkela, INDIA*
National Institute of Technology, Rourkela Apr. 2014 – May 2019

SKILLS

Technical Skills: Optical characterisation, Nanofabrication, Spectroscopic analysis, Experimental design, Electronic Instrumentation, Working in Cleanrooms, Cryogenics, Electron Microscopy(SEM), AFM, COMSOL.
Applications: Python (numpy, pandas, matplotlib, Seaborn), OriginLabs, C, C++, Latex, Fortran.
Language: English (C1), French (A1).
Leadership Skills:

- Managed and trained master students/new users on optical setup.

- Trained new users on transfer and fabrication of heterostructures.
- Planned and organised yearly meetings to increase group integration.

PUBLICATIONS

1. **Approaching the Intrinsic Properties of Moiré Structures Using Atomic Force Microscopy(AFM) Ironing.**
   Palai, S. K., Dyksik, M., Sokolowski, N. . . & Plochocka, P. (2023). *Nano Letters.*
2. **Twist-angle dependent dehybridization of momentum-indirect excitons in $MoSe_2/MoS_2$ heterostructures.**
   Sokołowski, N., Palai, S., Dyksik, M., Posmyk, K., Baranowskki, M., Surrente, A., . . . & Plochocka, P. (2023). *2D Materials.*
3. **Control of the Valley Polarization of Monolayer WSe_2 by Dexter-like Coupling.**
   Jasinski, J., Thompson, J., Palai, S., Smiertka, M.,. . . & Plochocka, P. (2024). *2D Materials.*
4. **Low-Temperature Disorder and Lattice Dynamics in a Luminescent 1D Hybrid Halide: Implications and Insights.**
   Andrzej Nowok, Mirosław Mączka, Anna Gągor, Maciej Ptak, Jan Zaręba, Swaroop Palai, Adam Sieradzki (2024). *ACS Chemistry of Materials.*
5. **Spin injection and emission helicity switching in a 2D perovskite/WSe_2 heterostructure.**
   Jasinski, J., Gucci, F., Brumme, T., Palai, S., Genco, A. . . & Surrente, A. (2026). *Nano Letters.*
6. **Directional emission from nitrogen-vacancy centers deterministically coupled to inverse-designed silicon nanoantennas.**
   Palai, S. K., Azéma, A., Müller, J., Hernandez, R., Agez, G., Poumirol, J.-M., Paillard, V., Larrieu, G., Wiecha, P. R., Ressier, L., & Cuche, A. (Communicated to Nanotechnology)
7. **Bimolecular momentum-dark exciton annihilation in atomically thin MoS_2.**
   Palai, S. K., Lamsaadi, H., Katsipoulaki, E., Shekhawat, V., Oualla, H., Agez, G., Groenen, J., Feddi, E., Chaudhary, V., Stratakis, E., Balocchi, A., Paradisanos, I., Combe, N., Cuche, A., Paillard, V., & Poumirol, J.-M. (Communicated to Applied Physics Letters)
8. **Probing spin-dark excitons via room-temperature photoluminescence in hBN-encapsulated monolayer MoS_2.**
   Lamsaadi, H., Palai, S. K., Katsipoulaki, E., Oualla, H., Agez, G., Groenen, J., Feddi, E., Chaudhary, V., Stratakis, E., Paradisanos, I., Cuche, A., Paillard, V., & Poumirol, J.-M. (Manuscript under preparation)

AWARDS AND FELLOWSHIPS

- NanoX Postdoctoral Fellowship.
- ANR Postdoctoral Fellowship.
- Marie Skłodowska-Curie Actions early stage researcher fellowship.
- EMFL proposal grant.
- Laserlab Europe Grant 2023.
- Junior Research Fellowship, IIT Madras, India.
- GATE 2018 qualified.
- Gold Medalist in the National Science Olympiad.

SCIENTIFIC COMMUNICATION

- Invited talk at META 2025, Torremolinos, Malaga, Spain.
- Invited talk at AES 2025, Tangier, Morocco.
- Talk at Flatlands 2023: Beyond Graphene, Prague, Czech Republic.
- 3-minute flash talk at Flatlands 2022: Beyond Graphene, Lanzarote, Canary Islands.
- Invited talk at CNRS-LNCMI Grenoble and Toulouse, June 2023.
- Poster presentation at EMFL days, Kerkerde, Netherlands.
- Poster presentation at EPFL, Lausanne, Switzerland.
- Poster presentation at CSIC-ICMM, Madrid, Spain.

WORKSHOPS

- Spring School on Scientific Communication, POLIMI, Milan, Italy.
- Summer School on Device Fabrication and Characterisation, EPFL, Lausanne, Switzerland.
- Winter School on van der Waal Materials, CSIC-ICMM, Madrid, Spain.

RELEVANT COURSEWORK

- Nanomaterial Maker's Course - Duke University (certificate)
- Semiconductor Packaging - Arizona State University (certificate)
- Thin film Characterisation and Ellipsometry - Paul Sabatier Toulouse III University.

HOBBIES

Photography, Video editing, Pasta making, Badminton, Swimming, Surfing, Patzer.

REFEREES

- **Paulina Plochocka** (Supervisor) - paulina.plochocka@lncmi.cnrs.fr
- **Andres Castellanos Gomez** (Intern supervisor) - andres.castellanos@csic.es
- **Carmen Munuera** (Intern Supervisor) - cmunuera@icmm.csic.es
- **Aurelien Cuche** (Postdoc Supervisor) - aurelien.cuche@cemes.fr
- **Laurence Ressier** (Postdoc Supervisor) - laurence.ressier@insa-toulouse.fr
- **Alessandro Surrente** (Mentor) - alessandro.surrente@pwr.edu.pl
- **Mateusz Dyksik** (Mentor) - mateusz.dyksik@pwr.edu.pl
- **Bamadev Das** (Mentor) - bamadev.das@ugent.be